ZCAS University

CCS1601 INTERNET OF THINGS FUNDAMENTALS

FINAL EXAM

2ND JUNE 2025

Time: 14:0 – 17:00 Hours

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

"Smart Home System "Home Ease"

You have been hired as a systems analyst for **TechSmart Solutions**, a company that designs intelligent home automation systems. One of their flagship products is **HomeEase**, a smart home platform that allows users to control various home devices such as lights, security cameras, thermostats, door locks, and appliances

using voice commands, a mobile app, or automated scheduling.

HomeEase uses:

- **IoT (Internet of Things)** devices to collect and transmit data.
- A **central AI assistant** to process user commands and adapt preferences.
- **Cloud-based services** for data storage and remote access.
- **Security features** like biometric access, end-to-end encryption, and anomaly detection.

The system can learn patterns of behavior, such as when lights should turn on/off, preferred room temperatures, and commonly used appliances. It can also detect irregular activities, such as unexpected entries or forgotten appliances left on, and notify users through mobile alerts.

Recently, the company is considering enhancing **HomeEase** with:

- Integration with **blockchain** for secure access logs.
- More advanced **predictive maintenance** features for connected appliances.
- **Interoperability with third-party smart systems** like Google Home and Alexa.

a) Define a **Smart Home System** and explain how **IoT** enables smart home automation (5 marks)

b) Identify and explain any **four components or technologies** involved in the Home Ease system. (8 marks)

c) Discuss **three benefits** and **two challenges** of using AI in the Home Ease smart home system. (5 marks)

d) Suppose you were tasked to collect user requirements for upgrading Home Ease. List **four questions** you would ask users to gather relevant requirements. (4 marks)

e) Write a python program that asks the user to enter their age. If the age is between 18 and 25 (inclusive) print "Young Adult". If the age is less than 18 print "Minor". If the age is above 25 print "Adults". (5 marks)

f) Numbers = [2, 4, 6, 8, 10, 12].

Write a python program that asks a user to enter a number. Check if the number is in the list. If it is print "Found". If not append the number to the list and print the updated list. (4 marks)

g) Colors = ("red", "green", "yellow", "blue") Write a python code that will ask the user to input a color. If the color exists in the tuple print the index of the color in the tuple. If it doesn't exist, Print "Color not in the tuple". (4 marks)

h) Students = {"Andy":85, "Mark":90, "Sue":50, "Rose":76} Ask the user to enter a student name. If the student is in the dictionary, check the score.

If its 90 and above print "Excellent"

If its between 70 and 89 print "Very Good"

If its 50 and below print "Needs Improvement"

(5 marks)

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

SmartMart, a growing retail chain, has recently implemented an IoT-based smart inventory and customer experience system across its stores. Using RFID tags, smart shelves, and IoT sensors, SmartMart can track inventory levels in real time, automatically reorder stock, and monitor product movement. Additionally, in-store sensors collect data on customer movement and product interactions to optimize shelf layouts and personalized promotions.

The system is integrated with a mobile app that notifies customers of discounts based on their preferences and previous purchases. SmartMart's management is evaluating how this IoT system has impacted operations, customer satisfaction, and business intelligence.

a) Define **IoT** and explain its role in modern business operations. (4 marks)

b) Based on the scenario, identify and explain **three specific IoT applications** used by SmartMart. (6 marks)

c) Discuss **Two business benefits** and **one potential challenge** of using IoT in SmartMart. (5 marks)

d) Suggest **Two ways** SmartMart could use IoT data to gain a **competitive advantage** (5 marks)

(20 marks)

Question 3

A university research team is developing an **automated greenhouse system** that maintains optimal conditions for plant growth. The system uses **sensors** to measure temperature, humidity, and soil moisture levels. Based on this input, the system controls **actuators** such as fans, irrigation pumps, and window servos. A **microcontroller** is programmed to process sensor data and trigger the appropriate actuators in real time, ensuring plants receive the right environment 24/7 without human intervention.

(a) Define the terms **sensor**, **actuator**, and **microcontroller**, and briefly describe how each is used in the greenhouse system. (6 marks)

(b) Identify **two types of sensors** that would be useful in the greenhouse system and explain the purpose of each.

(4 marks)

(c) Describe the **role of the microcontroller** in coordinating the actions of sensors and actuators in the system. (4 marks)

(d) List and explain **three advantages** of automating the greenhouse system using sensors, actuators, and a microcontroller. (6 marks)

(20marks)

Question 4

SmartAgro is a modern agricultural company that uses IoT devices to monitor soil moisture, weather conditions, and crop health in real time. These devices send data to nearby fog nodes deployed on-site for preliminary processing and decision-making (e.g., activating irrigation). The processed data is then sent to a centralized cloud platform for storage, further analysis, and visualization.

The company's objective is to improve crop yield while minimizing water usage. The system must respond quickly to changing environmental conditions and provide long-term analytics and forecasting.

(a) Differentiate between **Cloud Computing** and **Fog Computing** and explain why SmartAgro needs both. (5 marks)

(b) Describe two benefits and one limitation of using fog computing in SmartAgro's system. (5 marks)

(c) Explain how **network latency** affects real-time agricultural decision-making in SmartAgro, and how fog computing helps mitigate this issue. (5 marks)

(d) Identify two network components that are essential for SmartAgro's system and explain their role. (5 marks)

(20 marks)

Question 5

In today's digital age, software plays a central role in nearly every aspect of life. From health monitoring apps and smart home systems to automated banking platforms and national security infrastructure, software powers the tools we depend on. The COVID-19 pandemic accelerated the adoption of software-driven solutions in healthcare, education, and remote work. Organizations now face the challenge of ensuring that their software is secure, reliable, and adaptable in a rapidly changing world.

a) Explain the phrase **"Software is Everything"** in the context of modern society. (4 marks)

b) Identify and briefly explain **three major categories of software**, giving one example of each. (6 marks)

c) Discuss **two ways** software has transformed the **healthcare or education** sector. (4 marks)

d) List and briefly describe **three key qualities of good software** that organizations must prioritize today. (6 marks)

(20 marks)

END OF EXAM